# Mathematics for Social Scientists

Extra Exercises: OLS in Matrix Form

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Throughout, consider the linear regression model

$$y = X\beta + \varepsilon,$$

where $y$ is an $n \times 1$ vector of outcomes, $X$ is an $n \times (k+1)$ design matrix (with an intercept), $\beta$ is a $(k+1) \times 1$ parameter vector, and $\varepsilon$ is the error term. The OLS estimator is

$$\hat{\beta} = (X'X)^{-1}X'y,$$

provided that $X$ has full column rank so that $X'X$ is invertible.

The exercises below ask you to *compute* $\hat{\beta}$ "by hand", using matrix multiplication and matrix inversion (for $2 \times 2$ and $3 \times 3$ matrices).

## Exercise 1

You observe $n = 3$ units and estimate a simple linear regression with an intercept and one regressor ($k = 1$):

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, 2, 3.$$

In matrix form, the data are

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}.$$

(a) Write the model in matrix form by identifying $y$, $X$ and $\beta$ in the equation

$$y = X\beta + \varepsilon.$$

What are the dimensions of each object?

(b) Compute $X'X$ and $X'y$ explicitly.

(c) Recall that for a generic $2 \times 2$ matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det(A) = ad - bc,$$

and if $\det(A) \neq 0$, then

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Using this formula, compute $(X'X)^{-1}$.

(d) Compute the OLS estimator

$$\hat{\beta} = (X'X)^{-1}X'y$$

by performing the matrix multiplication. Write down the numerical values of $\hat{\beta}_0$ and $\hat{\beta}_1$.

(e) Interpret $\hat{\beta}_0$ and $\hat{\beta}_1$ in words (intercept and slope).

## Exercise 2

Now consider a multiple regression with an intercept and two regressors ($k = 2$) for $n = 4$ observations:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + u_i, \quad i = 1, \ldots, 4.$$

In matrix form, the data are

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 \\ 3 \\ 4 \\ 7 \end{bmatrix}.$$

(a) Compute $X'X$ and $X'y$ explicitly. Check that $X'X$ is symmetric.

(b) Compute the inverse of $A = X'X$, i.e. find $(X'X)^{-1}$.

(c) Compute

$$\hat{\beta} = (X'X)^{-1}X'y.$$

Write down the numerical values of $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$.

(d) Briefly interpret the three coefficients in the regression

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \hat{\beta}_2 X_{i2}.$$

**For reference:**

- *OLS estimator in matrix form:*

$$\hat{\beta} = (X'X)^{-1}X'y \quad \text{if } X'X \text{ is invertible.}$$

- $2 \times 2$ *inverse:* For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $ad - bc \neq 0$,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

- $3 \times 3$ *inverse via adjugate:* For $A \in \mathbb{R}^{3\times 3}$,

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A), \qquad \operatorname{adj}(A) = C^{\top},$$

where $C$ is the cofactor matrix with entries $C_{ij} = (-1)^{i+j} M_{ij}$.

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Solutions: Extra Exercises

## Exercise 1 (Solution).

(a) In matrix form,

$$y = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \qquad \varepsilon = \begin{bmatrix} \varepsilon_0 \\ \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}.$$

Dimensions:

$$X : 3 \times 2, \quad y : 3 \times 1, \quad \beta : 2 \times 1, \quad \varepsilon : 3 \times 1.$$

(b) Compute $X'X$ and $X'y$.

First,

$$X' = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}, \quad X'X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}.$$

Carry out the multiplication:

$$X'X = \begin{bmatrix} 1^2 + 1^2 + 1^2 & 1 \cdot 1 + 1 \cdot 2 + 1 \cdot 3 \\ 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 1 & 1^2 + 2^2 + 3^2 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}.$$

Next,

$$X'y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 + 3 + 5 \\ 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 23 \end{bmatrix}.$$

(c) Invert $X'X$ using the $2 \times 2$ formula. Here

$$A = X'X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad \det(A) = 3 \cdot 14 - 6 \cdot 6 = 42 - 36 = 6 \neq 0.$$

Therefore

$$(X'X)^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}.$$

(d) Compute

$$\hat{\beta} = (X'X)^{-1} X'y = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix}.$$

First multiply:

$$\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 10 \\ 23 \end{bmatrix} = \begin{bmatrix} 14 \cdot 10 - 6 \cdot 23 \\ -6 \cdot 10 + 3 \cdot 23 \end{bmatrix} = \begin{bmatrix} 140 - 138 \\ -60 + 69 \end{bmatrix} = \begin{bmatrix} 2 \\ 9 \end{bmatrix}.$$

Then divide by 6:

$$\hat{\beta} = \frac{1}{6} \begin{bmatrix} 2 \\ 9 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{3}{2} \end{bmatrix}.$$

So

$$\boxed{\hat{\beta}_0 = \frac{1}{3}, \quad \hat{\beta}_1 = \frac{3}{2}.}$$

(e) Interpretation:

- $\hat{\beta}_0 = \dfrac{1}{3}$ is the estimated intercept: the predicted value of $Y$ when $X = 0$.

- $\hat{\beta}_1 = \dfrac{3}{2}$ is the estimated slope: increasing $X$ by 1 unit increases the predicted $Y$ by $1.5$ units.

**Exercise 2 (Solution).**

(a) Compute $X'X$ and $X'y$.

We have

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \end{bmatrix}, \qquad X' = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Then

$$X'X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \\ 1 & 3 & 0 \end{bmatrix}.$$

Compute entrywise:

$$X'X = \begin{bmatrix} 1^2+1^2+1^2+1^2 & 0\cdot 1+1\cdot 1+2\cdot 2+3\cdot 3 & 1\cdot 1+1\cdot 0+1\cdot 1+1\cdot 0 \\ \cdots & 0^2+1^2+2^2+3^2 & \cdots \\ \cdots & \cdots & 1^2+0^2+1^2+0^2 \end{bmatrix} = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 14 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

(You can fill in the symmetric entries to check.)

Similarly,

$$X'y = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 0+3+4+7 \\ 0\cdot 0+1\cdot 3+2\cdot 4+3\cdot 7 \\ 1\cdot 0+0\cdot 3+1\cdot 4+0\cdot 7 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \\ 4 \end{bmatrix}.$$

Note that $X'X$ is symmetric (as expected).

(b)–(e) Invert $A = X'X$ using cofactors/adjugate and compute $\det(A)$.

Let

$$A = X'X = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 14 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

We use the adjugate formula

$$A^{-1} = \frac{1}{\det(A)}\,\mathrm{adj}(A), \quad \text{where } \mathrm{adj}(A) = C',$$

and $C = [C_{ij}]$ is the cofactor matrix, $C_{ij} = (-1)^{i+j} M_{ij}$, with $M_{ij}$ the minor.

*Minors and cofactors.* To get a minor $M_{ij}$, delete row $i$ and column $j$ from $A$ and take the determinant of the remaining $2 \times 2$ matrix. For example, $M_{11}$ comes from deleting the first row and first column:

$$M_{11} = \det \begin{bmatrix} 14 & 2 \\ 2 & 2 \end{bmatrix}.$$

Then attach the sign $(-1)^{i+j}$ to get the cofactor $C_{ij}$. The sign pattern is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Compute each $2 \times 2$ determinant:

$$\begin{aligned}
M_{11} &= \det \begin{bmatrix} 14 & 2 \\ 2 & 2 \end{bmatrix} = 14 \cdot 2 - 2 \cdot 2 = 28 - 4 = 24, && C_{11} = +24, \\
M_{12} &= \det \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix} = 6 \cdot 2 - 2 \cdot 2 = 12 - 4 = 8, && C_{12} = -8, \\
M_{13} &= \det \begin{bmatrix} 6 & 14 \\ 2 & 2 \end{bmatrix} = 6 \cdot 2 - 14 \cdot 2 = 12 - 28 = -16, && C_{13} = -16, \\
M_{21} &= \det \begin{bmatrix} 6 & 2 \\ 2 & 2 \end{bmatrix} = 12 - 4 = 8, && C_{21} = -8, \\
M_{22} &= \det \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} = 4 \cdot 2 - 2 \cdot 2 = 8 - 4 = 4, && C_{22} = +4, \\
M_{23} &= \det \begin{bmatrix} 4 & 6 \\ 2 & 2 \end{bmatrix} = 4 \cdot 2 - 6 \cdot 2 = 8 - 12 = -4, && C_{23} = +4, \\
M_{31} &= \det \begin{bmatrix} 6 & 2 \\ 14 & 2 \end{bmatrix} = 6 \cdot 2 - 14 \cdot 2 = 12 - 28 = -16, && C_{31} = -16, \\
M_{32} &= \det \begin{bmatrix} 4 & 2 \\ 6 & 2 \end{bmatrix} = 4 \cdot 2 - 6 \cdot 2 = 8 - 12 = -4, && C_{32} = +4, \\
M_{33} &= \det \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix} = 4 \cdot 14 - 6 \cdot 6 = 56 - 36 = 20, && C_{33} = 20.
\end{aligned}$$

Thus the cofactor matrix and adjugate are

$$C = \begin{bmatrix} 24 & -8 & -16 \\ -8 & 4 & 4 \\ -16 & 4 & 20 \end{bmatrix}, \qquad \operatorname{adj}(A) = C' = \begin{bmatrix} 24 & -8 & -16 \\ -8 & 4 & 4 \\ -16 & 4 & 20 \end{bmatrix}.$$

*Determinant (Sarrus' rule).* For the $3 \times 3$ matrix

$$A = \begin{bmatrix} 4 & 6 & 2 \\ 6 & 14 & 2 \\ 2 & 2 & 2 \end{bmatrix},$$

Sarrus' rule gives

$$\det(A) = (4 \cdot 14 \cdot 2 + 6 \cdot 2 \cdot 2 + 2 \cdot 6 \cdot 2) - (2 \cdot 14 \cdot 2 + 2 \cdot 2 \cdot 4 + 2 \cdot 6 \cdot 6).$$

Compute:

$$4 \cdot 14 \cdot 2 = 112, \quad 6 \cdot 2 \cdot 2 = 24, \quad 2 \cdot 6 \cdot 2 = 24 \ \Rightarrow\ 112 + 24 + 24 = 160,$$

$$2 \cdot 14 \cdot 2 = 56, \quad 2 \cdot 2 \cdot 4 = 16, \quad 2 \cdot 6 \cdot 6 = 72 \ \Rightarrow\ 56 + 16 + 72 = 144.$$

Hence

$$\det(A) = 160 - 144 = 16.$$

Therefore

$$A^{-1} = \frac{1}{\det(A)}\,\mathrm{adj}(A) = \frac{1}{16}\begin{bmatrix} 24 & -8 & -16 \\ -8 & 4 & 4 \\ -16 & 4 & 20 \end{bmatrix}.$$

Equivalently, in simplified fractions:

$$(X'X)^{-1} = \begin{bmatrix} \frac{24}{16} & -\frac{8}{16} & -\frac{16}{16} \\ -\frac{8}{16} & \frac{4}{16} & \frac{4}{16} \\ -\frac{16}{16} & \frac{4}{16} & \frac{20}{16} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} & -\frac{1}{2} & -1 \\ -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ -1 & \frac{1}{4} & \frac{5}{4} \end{bmatrix}.$$

This answers parts (b)–(e).

*Alternative method (Gauss–Jordan).* As a second way to invert $A$, we can use row operations on the augmented matrix $[A \mid I]$:

$$\left[\begin{array}{ccc|ccc} 4 & 6 & 2 & 1 & 0 & 0 \\ 6 & 14 & 2 & 0 & 1 & 0 \\ 2 & 2 & 2 & 0 & 0 & 1 \end{array}\right].$$

Step 1. Scale the first row and eliminate below:

$$R_1 \leftarrow \tfrac{1}{4}R_1, \quad R_2 \leftarrow R_2 - 6R_1, \quad R_3 \leftarrow R_3 - 2R_1.$$

This gives

$$\left[\begin{array}{ccc|ccc} 1 & \frac{3}{2} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 5 & -1 & -\frac{3}{2} & 1 & 0 \\ 0 & -1 & 1 & -\frac{1}{2} & 0 & 1 \end{array}\right].$$

Step 2. Create a leading 1 in the second row and eliminate above/below:

$$R_2 \leftarrow \tfrac{1}{5}R_2, \quad R_1 \leftarrow R_1 - \tfrac{3}{2}R_2, \quad R_3 \leftarrow R_3 + R_2.$$

We obtain

$$\left[\begin{array}{ccc|ccc} 1 & 0 & \frac{4}{5} & \frac{7}{10} & -\frac{3}{10} & 0 \\ 0 & 1 & -\frac{1}{5} & -\frac{3}{10} & \frac{1}{5} & 0 \\ 0 & 0 & \frac{4}{5} & -\frac{4}{5} & \frac{1}{5} & 1 \end{array}\right].$$

Step 3. Create a leading 1 in the third row and eliminate above:

$$R_3 \leftarrow \tfrac{5}{4}R_3, \quad R_1 \leftarrow R_1 - \tfrac{4}{5}R_3, \quad R_2 \leftarrow R_2 + \tfrac{1}{5}R_3.$$

This yields

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & -\frac{1}{2} & -1 \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & 1 & -1 & \frac{1}{4} & \frac{5}{4} \end{array}\right].$$

The right block is exactly $(X'X)^{-1}$ found above.

(f) Compute $\hat{\beta} = (X'X)^{-1}X'y$.

Recall that $A = X'X$ and

$$(X'X)^{-1} = A^{-1} = \frac{1}{16}\begin{bmatrix} 24 & -8 & -16 \\ -8 & 4 & 4 \\ -16 & 4 & 20 \end{bmatrix}, \qquad X'y = \begin{bmatrix} 14 \\ 32 \\ 4 \end{bmatrix}.$$

First compute the matrix–vector product

$$\operatorname{adj}(A)\, X'y = \begin{bmatrix} 24 & -8 & -16 \\ -8 & 4 & 4 \\ -16 & 4 & 20 \end{bmatrix} \begin{bmatrix} 14 \\ 32 \\ 4 \end{bmatrix} = \begin{bmatrix} 16 \\ 32 \\ -16 \end{bmatrix}.$$

Then divide by 16:

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{16}\begin{bmatrix} 16 \\ 32 \\ -16 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

So

$$\boxed{\hat{\beta}_0 = 1, \quad \hat{\beta}_1 = 2, \quad \hat{\beta}_2 = -1.}$$

(g) Interpretation:

- $\hat{\beta}_0 = 1$ is the estimated intercept: the predicted $Y_i$ when both regressors $X_{i1}$ and $X_{i2}$ are zero.
- $\hat{\beta}_1 = 2$ is the estimated effect of $X_{i1}$ on $Y_i$, holding $X_{i2}$ constant: increasing $X_{i1}$ by 1 increases the predicted $Y_i$ by 2 units.
- $\hat{\beta}_2 = -1$ is the estimated effect of $X_{i2}$ on $Y_i$, holding $X_{i1}$ constant: increasing $X_{i2}$ by 1 decreases the predicted $Y_i$ by 1 unit.